The Hermite Matching Matrix: A Combinatorial Change of Basis Between Hermite Polynomials and Hermite Functions

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Abstract

The Hermite polynomials $\{H_n(t)\}_{n\geq 0}$ are orthogonal with respect to the Gaussian weight e^{-t^2}. The Hermite functions $\psi_m(t) = c_m H_m(t)\, e^{-t^2/2}$ are obtained by absorbing the square root of this weight into the polynomial; they form an orthonormal basis for $L^2(\mathbb{R})$ and are eigenfunctions of the Fourier transform. The L^2-projection of the m-th Hermite function onto the Hermite polynomial basis—expressing ψ_m as $\sum_k a_{mk} H_k$—defines an infinite matrix (a_{mk}). We prove that, up to a single universal constant, every entry of this matrix is an integer, and we conjecture an explicit combinatorial formula: the rescaled entry D_{mk} counts *signed bipartite matchings* of $m+k$ vertices, where cross-pairs carry weight $+4$ and self-pairs carry weight -2. We give closed forms for the edge entries $D_{0,2n} = (-1)^n\,(2n)!/n!$ and show that the diagonal entries, divided by $n!$, are integers yielding the sequence $\sum_k \binom{n}{2k}\binom{2k}{k}\, 4^{n-2k}$ (OEIS A081671), with ordinary generating function $1/\sqrt{1-8x+12x^2}$ and exponential generating function $e^{4x} I_0(2x)$, where I_0 is the modified Bessel function of the first kind.

1 Setup

Let $H_n(t)$ denote the physicist's Hermite polynomials, defined by the recurrence

$$H_{n+1}(t) = 2t\, H_n(t) - 2n\, H_{n-1}(t), \qquad H_0(t) = 1, \quad H_1(t) = 2t,$$

and satisfying the orthogonality relation

$$\int_{-\infty}^{\infty} H_m(t)\, H_k(t)\, e^{-t^2}\, dt = 2^k\, k!\, \sqrt{\pi}\, \delta_{mk}. \tag{1}$$

The *Hermite functions* are

$$\psi_m(t) = c_m\, H_m(t)\, e^{-t^2/2}, \qquad c_m = (2^m\, m!\, \sqrt{\pi})^{-1/2}. \tag{2}$$

These form an orthonormal basis for $L^2(\mathbb{R})$ and are eigenfunctions of the Fourier transform with eigenvalues $(-i)^m$. Each ψ_m is entire, being a polynomial times a Gaussian.

The Hermite polynomials $\{H_k\}_{k\geq 0}$ form an orthogonal basis for $L^2(\mathbb{R}, e^{-t^2}\, dt)$. The best L^2-approximation of ψ_m by a polynomial of degree at most N in this basis is

$$\psi_m^{(N)}(t) = \sum_{k=0}^{N} a_{mk}\, H_k(t),$$

where

$$a_{mk} = \frac{c_m}{2^k\, k!\, \sqrt{\pi}} \int_{-\infty}^{\infty} H_m(t)\, H_k(t)\, e^{-3t^2/2}\, dt. \tag{3}$$

The factor $e^{-3t^2/2}$ arises from the product of $e^{-t^2/2}$ (from ψ_m) and e^{-t^2} (from the Gaussian weight defining the inner product). Our object of study is the infinite matrix (a_{mk}).

2 The Generating Function

Proposition 2.1. *The integrals $I_{mk} = \int_{-\infty}^{\infty} H_m(t)\, H_k(t)\, e^{-3t^2/2}\, dt$ have the generating function*

$$\sum_{m,k\geq 0} \frac{s^m\, u^k}{m!\, k!}\, I_{mk} = \sqrt{\frac{2\pi}{3}}\, \exp\Big(-\frac{s^2 - 4su + u^2}{3}\Big).$$

Proof. Using the generating function for Hermite polynomials, $\sum_{n\geq 0} H_n(t)\, z^n/n! = e^{2tz - z^2}$, we compute

$$\begin{aligned}
\sum_{m,k} \frac{s^m u^k}{m!\, k!}\, I_{mk} &= \int_{-\infty}^{\infty} e^{2ts - s^2}\, e^{2tu - u^2}\, e^{-3t^2/2}\, dt \\
&= \int_{-\infty}^{\infty} \exp\big(-\tfrac{3}{2}t^2 + 2(s+u)t - s^2 - u^2\big)\, dt \\
&= \sqrt{\frac{2\pi}{3}}\, \exp\Big(\frac{2(s+u)^2}{3} - s^2 - u^2\Big) \\
&= \sqrt{\frac{2\pi}{3}}\, \exp\Big(-\frac{s^2 - 4su + u^2}{3}\Big).
\end{aligned}$$

□

Corollary 2.2. *$I_{mk} = 0$ whenever $m + k$ is odd (parity).*

Proof. The exponent $-(s^2 - 4su + u^2)/3$ contains only terms of even total degree in (s, u). □

The matrix (a_{mk}) factors as

$$a_{mk} = R_{mk} \cdot \frac{\sqrt{2/3}}{\pi^{1/4}}, \tag{4}$$

where R_{mk} is algebraic (involving only rationals and square roots), and $\sqrt{2/3}/\pi^{1/4}$ is a universal constant independent of m and k.

3 The Rescaled Integer Matrix

Extracting coefficients from Proposition 2.1 yields the following.

Proposition 3.1. *Define*

$$D_{mk} = 3^{(m+k)/2} \cdot \frac{I_{mk}}{\sqrt{2\pi/3}}. \tag{5}$$

Then D_{mk} is an integer for all $m, k \geq 0$ with $m + k$ even (and zero when $m + k$ is odd).

Proof. Write

$$\exp\Big(-\frac{s^2 - 4su + u^2}{3}\Big) = \exp\Big(-\frac{s^2}{3}\Big)\, \exp\Big(\frac{4su}{3}\Big)\, \exp\Big(-\frac{u^2}{3}\Big).$$

The coefficient of $s^m u^k$ is a sum over triples (a, b, c) with $m = 2a + b$ and $k = b + 2c$:

$$[s^m u^k] = \sum_{a,b,c} \frac{(-1/3)^a}{a!}\, \frac{(4/3)^b}{b!}\, \frac{(-1/3)^c}{c!}.$$

Since $I_{mk} = m!\, k!\, [s^m u^k] \cdot \sqrt{2\pi/3}$, we have $D_{mk} = 3^{(m+k)/2} \cdot m!\, k! \cdot [s^m u^k]$. Writing $a + c = (m + k - 2b)/2$ and noting $3^{(m+k)/2}$ absorbs all powers of 3 from the denominators while $4^b = 2^{2b}$ and $(-1)^{a+c}$ provide integer factors, every term is an integer. □

4 Apparent Matching Interpretation

The integers D_{mk} appear to count signed bipartite matchings. We have verified the following against the proved coefficient-extraction formula of Proposition 3.1 for every pair (m, k) with $0 \le m, k \le 20$ (441 entries, zero discrepancies; see Appendix A).

Observation 4.1. *For all computed cases, D_{mk} equals the sum over all perfect matchings of $m + k$ vertices—m of type A and k of type B—where each matched pair contributes:*

Pair type	*Raw weight*	*Rescaled weight*
Cross (A–B)	$4/3$	$+4$
Self (A–A or B–B)	$-2/3$	-2

Rescaled weights are obtained by multiplying through by $3^{(m+k)/2}$. In closed form, this matching count would give

$$D_{mk} = \sum_{\substack{c \ge 0 \\ m-c \text{ even} \\ k-c \text{ even}}} \binom{m}{c}\binom{k}{c}\, c!\, (m-c-1)!!\, (k-c-1)!!\, 4^c\, (-2)^{(m+k-2c)/2}, \tag{6}$$

where $(2r-1)!! = 1 \cdot 3 \cdot 5 \cdots (2r-1)$ is the double factorial and $(-1)!! = 1$ by convention. The factors count: $\binom{m}{c}\binom{k}{c}$ ways to choose which vertices cross-pair, $c!$ ways to match them (each cross-pair weighted $+4$), and $(m-c-1)!!\,(k-c-1)!!$ self-matchings of the remainder (each self-pair weighted -2).

Remark 4.2. The matching interpretation is consistent with Wick's theorem applied to the expectation $\mathbb{E}[H_m(X)\, H_k(X)]$ for $X \sim N(0, 1/3)$. A proof would follow from Isserlis' theorem at non-unit variance.

5 The Diagonal

Theorem 5.1. *The diagonal entries satisfy*

$$\frac{D_{nn}}{n!} = \sum_{k=0}^{\lfloor n/2 \rfloor} \binom{n}{2k}\binom{2k}{k}\, 4^{n-2k}. \tag{7}$$

This is OEIS sequence A081671*, with ordinary generating function*

$$\sum_{n \ge 0} \frac{D_{nn}}{n!}\, x^n = \frac{1}{\sqrt{1 - 8x + 12x^2}} = \frac{1}{\sqrt{(1-2x)(1-6x)}}, \tag{8}$$

and exponential generating function

$$\sum_{n \ge 0} \frac{D_{nn}}{(n!)^2}\, x^n = e^{4x}\, I_0(2x), \tag{9}$$

where I_0 is the modified Bessel function of the first kind.

Proof. Setting $m = k = n$ in (5) and writing $j = (n-c)/2$ for the number of self-pairs per side:

$$D_{nn} = \sum_{j=0}^{\lfloor n/2 \rfloor} \binom{n}{n-2j}^2 (n-2j)!\, \big((2j-1)!!\big)^2\, 4^{n-2j}\, (-2)^{2j}.$$

Since $(-2)^{2j} = 4^j$ and $(2j-1)!! = (2j)!/(2^j\, j!)$, the summand becomes

$$\binom{n}{2j}^2 (n-2j)!\, \frac{(2j)!^2}{4^j\,(j!)^2}\, 4^j\, 4^{n-2j} = \binom{n}{2j}^2 (n-2j)!\, \frac{(2j)!^2}{(j!)^2}\, 4^{n-2j}.$$

Dividing by $n!$ and simplifying:

$$\frac{D_{nn}}{n!} = \sum_j \binom{n}{2j} \frac{(2j)!}{(j!)^2}\, 4^{n-2j} = \sum_j \binom{n}{2j}\binom{2j}{j} 4^{n-2j}.$$

For the OGF: let $a_n = \sum_k \binom{n}{2k}\binom{2k}{k} 4^{n-2k}$. Using $\sum_{n\geq 2k}\binom{n}{2k}(4x)^n = (4x)^{2k}/(1-4x)^{2k+1}$,

$$\sum_n a_n x^n = \sum_k \binom{2k}{k} 4^{-2k} \frac{(4x)^{2k}}{(1-4x)^{2k+1}} = \frac{1}{1-4x}\sum_k \binom{2k}{k}\Big(\frac{x}{1-4x}\Big)^{2k}.$$

Using the identity $\sum_k \binom{2k}{k} z^{2k} = 1/\sqrt{1-4z^2}$ with $z = x/(1-4x)$:

$$= \frac{1}{1-4x}\,\frac{1}{\sqrt{1-4x^2/(1-4x)^2}} = \frac{1}{\sqrt{(1-4x)^2-4x^2}} = \frac{1}{\sqrt{1-8x+12x^2}}.$$

For the EGF: $I_0(2x) = \sum_k x^{2k}/(k!)^2$, so $e^{4x}I_0(2x) = \sum_n\big(\sum_k \binom{n}{2k}\binom{2k}{k} 4^{n-2k}/n!\big)x^n$ by Cauchy product, confirming (9). □

Corollary 5.2 (Non-negativity)**.** *The diagonal entries $D_{nn}/n!$ are non-negative integers for all $n \geq 0$, being manifestly a sum of products of binomial coefficients.*

The first several values of $D_{nn}/n!$ are:

$$1,\ 4,\ 18,\ 88,\ 454,\ 2424,\ 13236,\ 73392,\ 411462,\ 2325976,\ \ldots$$

6 The Edge Entries

The boundary entries of the matrix have closed forms.

Proposition 6.1. *For $k = 2n \geq 0$,*

$$D_{0,2n} = (-1)^n \frac{(2n)!}{n!} = (-1)^n\, 2^n\, (2n-1)!!$$

These are, up to sign, the rising factorials $(2n)!/n!$.

Proof. When $m = 0$, the sum (5) (assuming its validity) has only the term $c = 0$, giving $D_{0,2n} = (2n-1)!!\,(-2)^n = (-1)^n(2n)!/n!$. Alternatively, direct extraction from the generating function: $[u^{2n}]\, e^{-u^2/3} = (-1/3)^n/n!$, so $I_{0,2n} = \sqrt{2\pi/3}\,(2n)!\,(-1/3)^n/n!$ and $D_{0,2n} = 3^n \cdot (2n)!\,(-1/3)^n/n! = (-1)^n(2n)!/n!$. □

7 Computed Values

The matrix D_{mk} is symmetric ($D_{mk} = D_{km}$), vanishes when $m+k$ is odd, and has alternating signs along each row. Since the even-indexed and odd-indexed entries decouple, we present them as two 8×8 blocks in Tables 1 and 2 (128 entries total). Table 3 lists $D_{nn}/n!$ for $0 \leq n \leq 30$. All entries were computed directly from the proved coefficient-extraction formula of Proposition 3.1. The matching formula (6) was independently checked against these values for all $0 \leq m, k \leq 20$ (441 entries, zero discrepancies).

8 Recovering the Original Matrix

The L^2-projection coefficients (3) are recovered from the integer matrix by

$$a_{mk} = \frac{D_{mk}}{3^{(m+k)/2}} \cdot \frac{1}{2^k\, k!\, \sqrt{2^m\, m!}} \cdot \frac{\sqrt{2/3}}{\pi^{1/4}}.$$

The first factor undoes the rescaling by $3^{(m+k)/2}$, the second accounts for the Hermite polynomial norms and the normalisation constant c_m, and the third is the universal constant.

9 Discussion

The change-of-basis matrix between Hermite functions and Hermite polynomials is defined by a continuous, analytic process: L^2-projection against a Gaussian weight. There is no *a priori* reason for the entries to be rational, let alone integer multiples of a single constant.

The integrality of D_{mk} (Proposition 3.1) is proved by direct coefficient extraction from the generating function. The matching interpretation (Observation 4.1) is verified computationally for all $m+k \leq 14$ but not proved here. A proof would likely follow from Wick's theorem (Isserlis' theorem) applied to the Gaussian expectation $\mathbb{E}[H_m(X)\, H_k(X)]$ for $X \sim N(0, 1/3)$.

The non-negativity of the diagonal $D_{nn}/n!$ is independent of the matching conjecture: it is a manifestly positive sum of products of binomial coefficients (Theorem 5.1), demanding a bijective interpretation.

The appearance of the modified Bessel function I_0 in the diagonal's EGF connects this matrix to wave equations and random walks. The OGF $1/\sqrt{(1-2x)(1-6x)}$ is an algebraic function—the simplest class beyond rational—with both branch points on the positive axis, consistent with the positivity of the diagonal and its growth rate 6^n. The matrix thus sits at an intersection of combinatorics (matchings), special functions (Bessel), algebraic generating functions, and classical analysis (Hermite, Fourier).

The matrix does not appear to have a standard name in the literature. We propose *Hermite matching matrix* for (D_{mk}) and note that a complete proof of the matching interpretation, a characterisation of the matrix's spectral properties, and its role in approximation theory remain open.

A Tables

Table 1: The integer matrix D_{mk} for even m and k, $0 \le m, k \le 14$.

$m\backslash k$	0	2	4	6	8	10	12	14
0	1	−2	12	−120	1680	−30240	665280	−17297280
2	−2	36	−408	6000	−110880	2479680	−65197440	1971889920
4	12	−408	10896	−254880	6471360	−184222080	5883736320	−209435466240
6	−120	6000	−254880	9529920	−339534720	12515731200	−491854809600	20808627840000
8	1680	−110880	6471360	−339534720	16590147840	−792232358400	38515400985600	−1948173273446400
10	−30240	2479680	−184222080	12515731200	−792232358400	48022189286400	−2872542897100800	173733606331084800
12	665280	−65197440	5883736320	−491854809600	38515400985600	−2872542897100800	208204121096294400	−14960314003822387200
14	−17297280	1971889920	−209435466240	20808627840000	−1948173273446400	173733606331084800	−14960314003822387200	1262574756680715878400

Table 2: The integer matrix D_{mk} for odd m and k, $1 \le m, k \le 15$.

$m\backslash k$	1	3	5	7	9	11	13	15
1	4	−24	240	−3360	60480	−1330560	34594560	−1037836800
3	−24	528	−9120	181440	−4233600	114428160	−3528645120	122464742400
5	240	−9120	290880	−8588160	266837760	−9021196800	334183449600	−13545845913600
7	−3360	181440	−8588160	369895680	−15425303040	658946534400	−29586651494400	1411067821363200
9	60480	−4233600	266837760	−15425303040	844050170880	−45401368320000	2476187275161600	−139467481548595200
11	−1330560	114428160	−9021196800	658946534400	−45401368320000	3021003695001600	−198959976152473600	13213366010836992000
13	34594560	−3528645120	334183449600	−29586651494400	2476187275161600	−198959976152473600	15600117724666675200	−1214739624489431040000
15	−1037836800	122464742400	−13545845913600	1411067821363200	−139467481548595200	13213366010836992000	−1214739624489431040000	109767194886547021824000

Table 3: The diagonal sequence $D_{nn}/n!$ for $0 \leq n \leq 30$. This is OEIS A081671, with OGF $1/\sqrt{1-8x+12x^2}$.

n	$D_{nn}/n!$	n	$D_{nn}/n!$
0	1	16	487610895942
1	4	17	2838118247064
2	18	18	16547996212044
3	88	19	96635257790352
4	454	20	565107853947444
5	2424	21	3308820294176016
6	13236	22	19395905063796312
7	73392	23	113814537122646432
8	411462	24	668494299227072796
9	2325976	25	3929851838287363824
10	13233628	26	23120826355480783896
11	75682512	27	136129681993195136032
12	434662684	28	802049367833112712312
13	2505229744	29	4728540783318487334496
14	14482673832	30	2789408149524132623521 6
15	83940771168		